

7(a)(i)

$$y = kMgl^r d^s b^{-1}$$

$$\ln y = r \ln l + s \ln d + \ln \left(\frac{kMg}{b} \right)$$

$$\ln y_1 = r \ln l + s \ln (5.00 \times 10^{-3}) + \ln \left(\frac{kMg}{b} \right) \text{----- (1)}$$

$$\ln y_2 = r \ln l + s \ln (6.00 \times 10^{-3}) + \ln \left(\frac{kMg}{b} \right) \text{----- (2)}$$

$$(1) - (2)$$

$$s = \frac{\ln \left(\frac{y_1}{y_2} \right)}{\ln \left(\frac{5}{6} \right)}$$

Using first row of data in Fig. 7.2,
 $s = -3$ (1 s.f.)

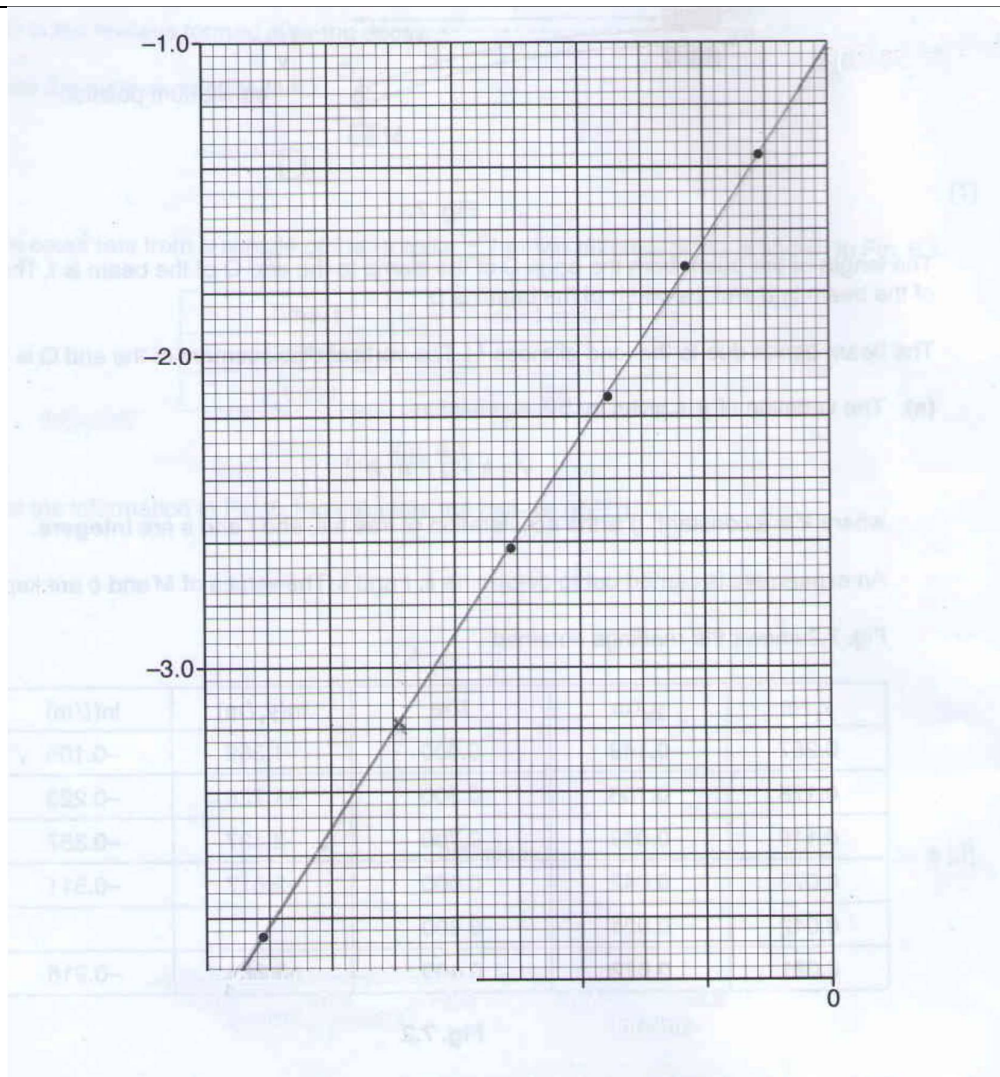
(a)(ii)

$$\ln(y_1 / \text{m}) = -3.170$$

$$\ln(l / \text{m}) = -0.693$$

(a)(iii)1.

(a)(iii)2.



Plot point (-0.693, -3.170) on Fig. 7.3

Sketch best fit line

(a)(iv)
$$\text{gradient} = \frac{-1.02 - (-4.00)}{0.00 - (-0.96)}$$
$$= 3.1$$

(a)(v) Using $y = mx + c$
Hence, $\ln y_1 = 3.1 \ln l - 1.02$

Fig. 7.3 shows a linear relationship between $\ln y_1$ and $\ln l$ with gradient 3.1 and vertical intercept -1.02. This can also be re-expressed as:

$$\ln y = r \ln l + s \ln d + \ln \left(\frac{kMg}{b} \right)$$

Where r and $s \ln d + \ln \left(\frac{kMg}{b} \right)$ are non zero. Hence graph in Fig. 7.3 supports the

(a)(vi) (b)	expression in (a). $r = 3$